

Towards Adversarial Collaboration in HCI Research

Oded Nov
Tandon School of Engineering
New York University
New York, USA
onov@nyu.edu

Bart Piet Knijnenburg
School of Computing
Clemson University
Clemson, South Carolina, USA
bartk@clemson.edu

Felicia Fang-Yi Tan
Tandon School of Engineering
New York University
New York, USA
felicia.tan@nyu.edu

Abstract

Human-Computer Interaction (HCI) research encompasses diverse and sometimes conflicting accounts of how and why people interact with technologies. We propose Adversarial Collaboration (AC) as a methodological framework for comparing and adjudicating conflicting HCI theories and design approaches. In AC, researchers holding divergent explanations and hypotheses jointly design and conduct experiments to adjudicate disagreements under mutually pre-agreed protocols. While AC has been used successfully in other scientific fields, it has not been systematically applied in HCI. We argue that HCI is well-positioned to benefit from AC, drawing on Boundary Object theory. We illustrate the potential value of AC through three experimental designs addressing ongoing HCI debates on privacy behavior, gamification and intrinsic motivation, as well as algorithmic transparency and trust. Together, these examples demonstrate how AC enables transparent, systematic comparison of competing HCI accounts, strengthening theoretical rigor and informing design practice.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; *Collaborative and social computing theory, concepts and paradigms*.

Keywords

Adversarial collaboration, Methodology, Scientific disagreement, Boundary objects, Open science

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1 Introduction

Human-Computer Interaction (HCI) research encompasses conflicting accounts of how and why people interact with technologies. This can lead to intractable debates and inconsistent theoretical approaches even when relevant data are available. For example, prior work has documented both algorithm aversion [6], where users abandon algorithms after observing errors despite superior accuracy [8], and algorithm appreciation, in which users overweigh

algorithmic advice relative to identical human advice [11, 20]. These opposing findings illustrate how HCI research can produce theoretical disagreement when studying closely related phenomena.

To address such discrepancies, we propose *Adversarial Collaboration* (AC) [5, 7, 16, 23] as a methodological innovation in HCI. In AC, researchers who hold opposing explanations articulate their positions, agree on research questions and hypotheses, design experiments to test these competing hypotheses, and publish results together [7]. ACs represent a form of team science that brings together diverse (and even contradictory) perspectives [5]. This approach curtails confirmation bias and post-hoc method revision: each side serves as a check on the other to ensure hypotheses are falsifiable and tests are agreed in advance. Originally applied in the cognitive and behavioral sciences (e.g., to adjudicate between competing models of judgment bias [22]), AC has since been applied across multiple social science domains [10, 13], with evidence that it enhances the reliability and validity of research findings [5].

AC can lead to reconciliation by clarifying the boundary conditions under which competing explanations apply. Boundary conditions refer to the specific circumstances under which a theoretical claim is expected to hold, including its limits of generalizability [2, 30]. For example, to reconcile the conflicting findings around algorithm aversion and appreciation, Hou and Jung [12] demonstrated that when an algorithm was described as more competent than a human peer, people showed algorithm appreciation, whereas framing the human as the authority produced algorithm aversion. Rather than assuming that one hypothesis must universally dominate, AC enables researchers to identify the task contexts, user characteristics, design framings, and measurement choices that systematically favor one account over another. In this paper, we demonstrate how AC can contribute to HCI in three ways:

- (1) **Methodological rigor and transparency:** AC enforces advance agreement on hypotheses, methods, and analyses, increasing transparency and reducing confirmation bias in line with open-science principles.
- (2) **Theoretical reconciliation:** By requiring jointly defined experiments, AC in HCI can neutralize partisan bias and test rival claims directly. This helps resolve contested debates by producing mutually agreed evidence, explain why prior studies diverged and identify replicable effects.
- (3) **Practical design implications:** In many HCI domains, design choices operationalize contested theoretical assumptions (e.g., whether explanations build or undermine trust, or whether gamification enhances or diminishes intrinsic motivation). By clarifying the mechanisms and boundary conditions under which competing accounts hold, AC enables designers to make deliberate decisions about which



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mechanisms to activate, calibrate, or avoid in specific contexts. This strengthens the foundation for design decisions in domains characterized by theoretical disagreement.

2 Studying Adversarial Collaboration using Boundary Objects

Durable tensions emerge when disjoint groups of scientists use different theoretical stances and methodological approaches. Such differences create a communication boundary that can be *syntactic* (a mismatch in knowledge representation, such as irreconcilable concept definitions), *semantic* (a mismatch in methods of knowledge creation, such as a lack of mutual recognition of research methodologies), and/or *pragmatic* (a mismatch in theoretical foundations that prevent findings from one side from transforming the other's knowledge) [4]. This communication boundary both exacerbates, and is reinforced by, the inherently tacit nature of knowledge within each group. Even when well documented, each group's scientific knowledge remains localized (presenting a limited view of the problem), embedded (difficult to understand without familiarity with the underlying research practice), and invested (hard-won and therefore resistant to fundamental change).

Here, we argue that the HCI community is uniquely positioned to apply AC and study its application, e.g., by applying boundary object theory. *Boundary objects* [4, 28] can help researchers cross this communication boundary by facilitating a process of *interesement*: knowledge transformation despite its localized, embedded, and invested nature [28]. Below, we draw on boundary object theory to facilitate interesement in the AC process through five components: maps, models, methods, tools, and repositories.

Governance Documents Mapping Out the Collaboration: In Boundary Object theory, “maps” are system-level representations that address the *pragmatic* communication boundaries by providing a shared vehicle for resolving conflict and transforming knowledge [4]. In our proposal, AC starts by drawing up governance documents that map the collaboration process. The documents specify: (1) the theoretical stakes of the collaboration, including definitions of the opposing positions, points of disagreement, and rival predictions they generate; (2) a study design for testing the predictions, including primary and secondary outcomes, and mutually agreed interpretations of possible results; (3) a collaboration plan detailing roles, conflict resolution procedures, data-sharing arrangements, and authorship policies; and (4) timelines, participant recruitment, preregistration, and analysis plans.

A Joint Research Model: “Models” play a role similar to maps, but operate at the level of collaborative practice. To avoid divergent interpretations of the research results, researchers create a joint model describing the rival predictions as hypothesized paths. This shared model makes competing theoretical assumptions and their empirical evaluation explicit within a single decisive study. Causal research models are particularly well-suited for this purpose, as they make mechanisms (mediating variables), boundary conditions (moderating variables), and the order of causality (path directions) explicit [18].

A Decisive Study as a Method to Evaluate the Research Model: Standardized methods address *semantic* communication boundaries by providing shared tools for explicating knowledge

that helps people learn about their differences and dependencies [4]. In our proposal, the decisive study serves this role by requiring opposing teams to agree upon a method for validating a joint research model. If methodological preferences cannot be reconciled, the decisive study can compare irreconcilable approaches against a mutually agreed-upon external yardstick. In HCI, this yardstick can be grounded in real-world impact, e.g., by evaluating whether attitudinal or behavioral trust measures better predict sustained human-AI collaboration.

Concept Definitions and Measurement Tools: “Tools” address *syntactic* communication boundaries by creating a shared language for common representations [4]. Key tools in AC include concept definitions and operationalizations agreed upon by researchers. If agreement on definitions or operationalizations cannot be reached, the decisive study can take the form of a measurement comparison, evaluating competing constructs in terms of predictive validity (prediction of an agreed outcome) and/or concurrent validity (convergence with an agreed related measure or differentiation between theoretically distinct groups).

A Repository of Materials and Outcomes: Repositories of materials and outcomes extend beyond individual collaborators to the research communities they represent, supporting the reintegration of conflicting findings. Similarly, jointly agreed concept definitions, measures and methods can be reused to align research practices across communities.

3 Adversarial collaboration in HCI: Illustrative examples

AC was developed in the social and behavioral sciences to address theoretical disputes, including disagreements over expert intuition [15], goal-setting theory [19], and income and well-being [16]. To date, however, AC has not been systematically applied within HCI. We therefore present three illustrative examples drawn from longstanding debates over conflicting explanations of user interaction, each outlining a potential AC-based experimental design (Table 2).

3.1 Privacy and Security Warnings

A persistent debate in HCI concerns why users routinely ignore security and privacy warnings [9, 14]. One account (H1) attributes this behavior to habituation: repeated exposure to similar warnings leads to attentional decay and automatic dismissal [1]. An opposing account (H2) attributes warning disregard to comprehension failure, arguing that technical or generic warning content prevents users from understanding the associated risks [25].

Proposed Experiment using the AC framework: Study participants perform web-browsing tasks (e.g., email) that trigger simulated security and privacy warnings under controlled presentation conditions. The opposing teams jointly design the warning conditions. The team emphasizing comprehension pushes for simplified, plain-language warnings, while the team emphasizing habituation ensures that those warnings are presented repetitively and uniformly to induce attentional decay. This joint manipulation addresses the *pragmatic* communication boundary by ensuring that teams commit ex ante to shared interpretation and analysis plan, and to accepting the observed outcome as adjudicating between

Boundary Object	Role in Adversarial Collaboration	Artifacts / Decisions
Maps: Governance Documents	Align expectations, incentives, and interpretations across opposing teams by stabilizing the collaboration process	Theoretical positions and rival predictions; agreed study design; roles and authorship; conflict resolution procedures; preregistration, timelines, ethics, and data licensing
Models: Joint Research Model	Make competing theoretical assumptions explicit within a shared causal framework that specifies mechanisms, boundary conditions, and hypothesized paths	Shared causal model specifying mediators, moderators, boundary conditions, and causal direction underlying rival explanations
Methods: Decisive Study	Provide a mutually agreed procedure for adjudicating between competing explanations or for comparing irreconcilable approaches against a shared criterion	Experimental or evaluative study design; agreed outcome criteria; external yardsticks such as real-world impact or predictive validity
Tools: Concept Definitions and Measures	Stabilize shared representations of constructs or enable systematic comparison when agreement cannot be reached	Joint construct definitions; agreed operationalizations; measurement comparison studies assessing predictive or concurrent validity
Repositories: Shared Research Materials	Support reuse, scrutiny, and community-level uptake of adversarial collaboration outcomes beyond the immediate collaboration	Open repositories of data, materials, models, measures, and analysis plans supporting cumulative research

Table 1: Boundary objects structuring adversarial collaboration in HCI.

the competing explanations. To address the *semantic* communication boundary, the competing explanations are formalized within a shared research model. This allows the competing mechanisms to be evaluated within a single, mutually agreed representational structure. Finally, to address the *syntactic* communication boundary, the teams agree in advance on a common set of measures, including warning compliance rate, response time, recall of warning content, and perceived understanding or annoyance, as well as on which measure(s) constitute the primary evaluation criterion in the event of divergent outcomes across multiple measures.

3.2 Gamification and Intrinsic Motivation

Another debate in HCI concerns whether gamification affects engagement by enhancing intrinsic motivation or by shifting users toward extrinsic regulation. One account (H1) argues that game elements such as points and badges increase engagement and intrinsic motivation by supporting psychological needs such as competence, autonomy, and relatedness [21]. In contrast, an opposing account (H2), grounded in Self-Determination Theory, suggests that extrinsic incentives may crowd out intrinsic motivation by shifting users' focus from the activity itself to external rewards [29].

Proposed Experiment using the AC framework: Participants perform a task designed to plausibly support intrinsic motivation (e.g., puzzle solving) using either a gamified interface or a non-gamified interface with identical core functionality. The opposing teams jointly co-design the interfaces. The team advocating gamification pushes for salient and richly implemented game elements to maximize motivational effects, while the opposing team ensures that the control interface is equally polished and that feedback remains informational. To address the *semantic* communication boundary, the competing accounts are formalized within a shared research model that distinguishes need satisfaction-based motivational mechanisms from reinforcement-driven behavioral effects. This model makes explicit how gamification is hypothesized to influence engagement either by enhancing intrinsic motivation or by introducing extrinsic control. To address the *syntactic* communication boundary, the teams agree on engagement metrics (e.g.,

task completion). Self-reported motivation and need satisfaction (e.g., perceived competence and autonomy) capture intrinsic motivational mechanisms, while behavioral indicators of reinforcement (e.g., rewards) capture extrinsic effects. By agreeing ex ante on how the measures map onto the competing theoretical accounts, the teams ensure that the outcomes adjudicate between the competing motivational mechanisms underlying engagement.

3.3 Algorithmic Transparency and Trust

A long-standing debate in HCI concerns how algorithmic transparency affects user trust [3, 17]. One account (H1) argues that explanations increase trust by improving users' understanding and acceptance of algorithmic decisions [17]; an opposing account (H2) suggests that transparency can undermine trust by exposing uncertainty or errors [27].

Proposed Experiment using the AC framework: In an online experiment, study participants perform an AI-assisted decision-making task (e.g., loan approval, content classification) using two interface variants: an *Explainable* condition that provides decision rationales and an *Opaque* condition presenting outcomes only. Rather than treating the conditions as fixed, the opposing teams jointly co-design the explanation interface. One team pushes for explanations that maximize clarity and usefulness, while the other ensures that explanations surface information that could reduce trust. This joint design process addresses the *pragmatic* communication boundary by constructing an interface that each side believes favors its own account, yielding opposing outcome predictions. The teams also agree ex ante on an analysis plan and commit to accepting either outcome as evidence supporting one account and rejecting the other, or the conditions under which either holds. These opposing predictions are formalized within a shared research model that specifies competing mediating mechanisms, such as perceived transparency versus perceived confidence in the system. This addresses the *semantic* communication boundary by making each side's explanatory assumptions explicit and comparable. Finally, the teams jointly agree on a common set of outcome measures, including trust and reliance, as well as on how effects should be interpreted. By

Example Domain	Competing Accounts	Experimental Manipulation	Measures
Privacy Warnings	H1: Habituation drives dismissal H2: Poor comprehension drives dismissal	Repeated uniform warnings vs. Varied/simplified warning content	Warning compliance; response time; recall
Gamification	H1: Gamification boosts motivation H2: Gamification undermines intrinsic motivation	Gamified interface vs. Non-gamified interface	Task engagement; continued use; motivation scale
Algorithmic Transparency	H1: Explanations increase trust H2: Explanations undermine trust	Explainable AI vs. Opaque AI	Trust-in-AI scale; reliance on AI recommendations

Table 2: Illustrative adversarial collaborations in HCI, showing how competing explanations are potentially adjudicated through jointly agreed manipulations and measures.

committing to shared operationalizations and interpretive criteria, the resulting data address the *syntactic* communication boundary and function as mutually acceptable evidence for adjudicating between the competing accounts.

4 Discussion

This paper positions Adversarial Collaboration (AC) as a response to persistent theoretical disagreements in HCI. Conflicting explanations of user behavior often lead to fragmented evidence and limited progress, even when empirical data are available. We argue that AC offers a way to reconcile such disagreements by requiring advance agreement on hypotheses, methods, and interpretations, without suppressing the pluralism of theories and practices that characterizes HCI. We further propose boundary object theory as a lens for analyzing and improving AC in HCI. By identifying governance documents, research models, methods, tools, and repositories as key boundary objects, we show how AC can bridge syntactic, semantic, and pragmatic communication boundaries between researchers with divergent theoretical views. The examples demonstrate how disagreements can be made explicit and empirically tractable, clarifying where competing accounts diverge.

Prior uses of AC suggest that it is most effective for empirical or theory-mediated empirical disagreements, where competing perspectives yield distinct, testable predictions that can be evaluated. The examples discussed here are representative of this class of disagreements. By contrast, AC is less applicable when disagreement cannot be translated into experimental comparisons. Accordingly, AC is ill-suited for early-stage or exploratory research that prioritizes generative exploration and theory building (e.g., ethnography). In addition, AC entails substantial coordination and labor costs. Prior accounts describe sustained negotiation across design, analysis, and interpretation phases [7], which can be particularly challenging in student-led projects or industry-academia collaborations where timelines and incentives diverge. These coordination demands are sensitive to power asymmetries among collaborators: differences in seniority or disciplinary authority can shape whose arguments carry more weight, even when participants are committed to epistemic neutrality [13]. Within HCI, where collaborations often span technical, design-oriented, and interpretive traditions, such asymmetries may be especially pronounced. Explicit governance structures are therefore critical, and boundary object theory provides a useful framework for designing shared artifacts that help surface assumptions and facilitate collaborations.

Despite these constraints, HCI is well positioned to adopt AC. The field integrates experimental psychology, systems engineering, design research, CSCW, and critical and participatory traditions, each operating under distinct evidentiary standards. As a result, disagreements often persist not because data are lacking, but because camps diverge in how constructs are defined and what counts as decisive evidence. Unlike many areas of cognitive and behavioral science, where disputes are typically framed as competing hypotheses evaluated through shared experimental paradigms and increasingly formalized open-science practices [24], HCI operates under heterogeneous contribution and evaluation criteria [26, 31]. This epistemic pluralism creates a strong need for structured mechanisms that surface and negotiate underlying assumptions. Empirical reflections suggest that AC rarely yields definitive resolutions. More often, it clarifies mechanisms, boundary conditions, and sources of prior divergence [13]. In HCI, where debates frequently concern the effects of interaction techniques or design interventions across contexts and populations, such clarification may be especially valuable. At the same time, the absence of unified adjudication norms and the prevalence of exploratory or artifact-centered work may complicate translating disagreements into mutually agreed decisive studies. Adopting AC in HCI would therefore require careful adaptation to its infrastructural diversity.

Beyond the types of disagreements illustrated here, future work can extend AC to disagreements over causal direction, where competing accounts posit opposing explanatory relationships between observed phenomena. Such disagreements are common in sociotechnical research, and an AC approach would require opposing teams to negotiate study designs that distinguish causal accounts, including agreement on constructs, measurement instruments, interventions, time horizons, and analytic criteria.

5 Conclusion

This work positions adversarial collaboration as a method for transforming persistent disagreement in HCI into shared empirical inquiry. By framing AC through boundary object theory, we show how advance agreement on models, methods, measures, and governance can bridge syntactic, semantic, and pragmatic divides. While AC is demanding and selectively applicable, it offers a principled way to make theoretical disagreement explicit, testable, and cumulative. We argue that adopting AC where conditions permit can strengthen HCI's ability to produce coordinated knowledge across its diverse theoretical and methodological traditions.

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